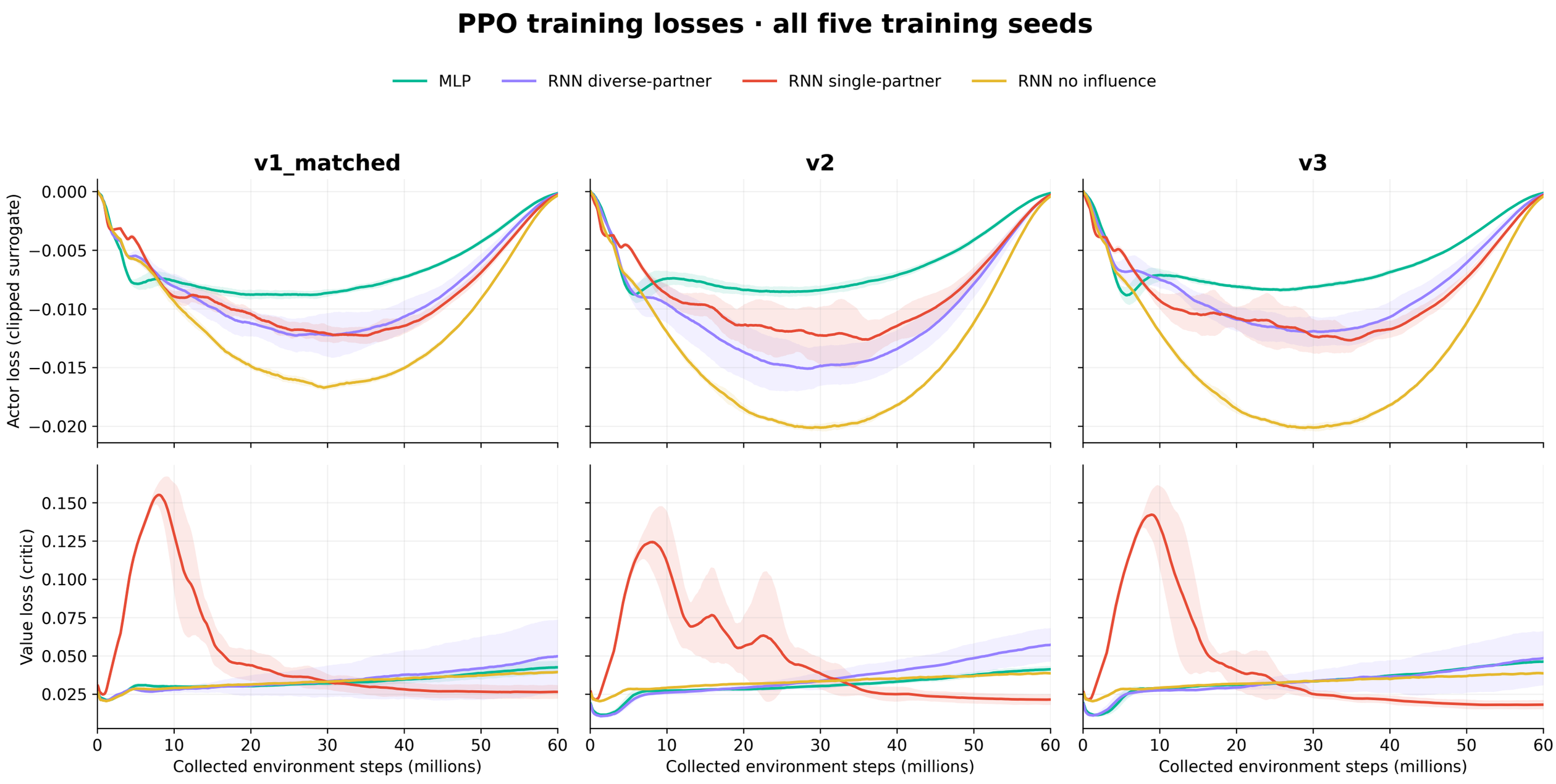

PPO training losses · all five training seeds
MLP
RNN diverse-partner
RNN single-partner
RNN no influence
v1_matched
v2
v3
Actor loss (clipped surrogate)
0.000
−0.005
−0.010
−0.015
−0.020
Value loss (critic)
0.150
0.125
0.100
0.075
0.050
0.025
0
10
20
30
40
50
60
Collected environment steps (millions)
Trailing 46-update mean (3.01M steps); early windows use available updates. Bands: sample SD across five seeds.